



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

AGRICULTURAL TECHNOLOGY

MAY/JUNE 2026

MARKING GUIDELINES

MARKS: 200

This marking guideline consists of 15 pages.

SECTION A**QUESTION 1**

1.1	1.1.1	C✓✓	
	1.1.2	C✓✓	
	1.1.3	B✓✓	
	1.1.4	D✓✓	
	1.1.5	B✓✓	
	1.1.6	A✓✓	
	1.1.7	C✓✓	
	1.1.8	C✓✓	
	1.1.9	A✓✓	
	1.1.10	A✓✓	(20)
1.2	1.2.1	Centre pivot✓✓	
	1.2.2	Universal joint/ Hooke's joint✓✓	
	1.2.3	Sieves✓✓	
	1.2.4	GPS✓✓	
	1.2.5	Galvanizing/✓✓	(10)
1.3	1.3.1	B✓✓	
	1.3.2	G✓✓	
	1.3.3	C✓✓	
	1.3.4	D✓✓	
	1.3.5	F✓✓	(10)

TOTAL SECTION A: 40

SECTION B**QUESTION 2: MATERIALS AND STRUCTURES**

Start this question on a NEW page.

- 2.1 2.1.1 **Alloy element that can be added to stainless steel to give it a fair amount of toughness.**
- Nickel/ Chromium✓ (1)
- 2.1.2 **Percentage of copper and zinc in this alloy.**
- 67% copper✓
 - 33% zinc✓ (2)
- 2.1.3 **Explanation of internal stress relieving in brass.**
- Heat the component to a temperature of 250–300 °C✓
 - Keep the component heated for 30 minutes to 1 hour✓
 - Leave on the side of the furnace to cool down slowly✓
 - Quench cracks may be caused if the component is too hot and cooled down too fast✓
 - Gives fairness/controlled cooling (4)
- 2.1.4 **Circumstances where bronze hammers can be used AND explanation.**
- Explosive atmospheres or in the presence of flammable vapours✓ because bronze when struck against a hard surface will not generate sparks.✓ (2)
- 2.1.5 **Explanation why bronze wool is preferred to steel wool for woodworking applications.**
- Steel wool will discolour oak/ Does not rust.✓ (1)
- 2.2 2.2.1 **TWO properties of an adhesive.**
- Water resistant✓
 - Non-conductive to electricity✓
 - Elastic✓
 - Heat resistant✓
 - Chemical resistant. ✓ (Any 2) (2)
- 2.2.2 **TWO ways to improve the cohesion properties of silicon.**
- Apply adhesive to both surfaces.✓
 - Too thick a layer of adhesive will result in a weak joint.✓
 - Apply a thin base coat if the surface is very porous.✓
 - Apply only a thin layer of adhesive.✓ (Any 2) (2)

- 2.3 2.3.1 **TWO reasons for using fiberglass instead of metal for the manufacturing of the cab.**
- Lightness✓
 - Water tight✓
 - Non-conductive of electricity✓
 - Can be formed into any shape✓
 - Easy to colour✓
 - Colour fast✓
 - Can be sawn, drilled, and filled✓
 - Toughness✓
 - Brittle when struck✓
 - Easily repaired when broken✓
- (Any 2) (2)
- 2.3.2 **TWO precautionary measures to follow when working with fibreglass.**
- Remove resin catalyst and accelerator from skin.✓
 - Wear gloves if skin is sensitive.✓
 - Use acetone in a well-ventilated room.✓
 - Handle resin casting carefully, they are brittle.✓
 - Glass fibre matting has small pieces that penetrate the skin.✓
 - Do not breathe in glass fibre or get it in your eyes.✓
 - Catalyst and accelerator should be stored separately to prevent explosions.✓
- (Any 2) (2)
- 2.3.3 **Steps to follow when repairing the crack on the cab.**
- Clean the surface.✓
 - Sand the surface.✓
 - Apply the resin and hardening agent.✓
- (3)
- 2.4 2.4.1 **TWO reasons for the use of Vesconite rather than white metal when manufacturing bushes.**
- Vesconite has a higher design load limit than white metal.✓
 - Vesconite has higher fatigue strength.✓
 - Vesconite has a much longer wear life.✓
 - Low friction.✓
 - Self-lubrication.✓
- (Any 2) (2)
- 2.4.2 **Explanation of why Vesconite is safe to use in the manufacturing industry.**
- Vesconite does not contain any asbestos.✓
 - Contains no lead.✓
 - No hazardous fibres or substances.✓
- (3)

2.4.3 TWO electrical properties of Teflon.

- The best sealant.✓
- Its dissipation factor is low, and yet Teflon has high dielectric strength.✓

(2)

2.4.4 THREE reasons why Vesconite is the best material for the manufacturing of bushes.

- It is ideal for many marine applications.✓
- It does not swell and seize.✓
- It remains hard.✓
- It does not delaminate.✓
- It has low friction.✓
- It does not corrode.✓
- Bushes can easily be removed.✓
- Does not contain any asbestos (healthy)✓/Safer to work with.
- Easy to machine.✓

(Any 3)

(3)

2.5 2.5.1 Component shown in the picture.

Isolator✓

(1)

2.5.2 TWO suitable materials for manufacturing an isolator.

- Plastic✓
- Porcelain✓

(2)

2.5.3 ONE property of the material identified in QUESTION 2.5.2.

Non-conductive to electricity✓

(1)

[35]

QUESTION 3: ENERGY

3.1 3.1.1 **TWO factors to consider before installing a wind turbine.**

- Surrounding environment.✓
- Efficiency.✓
- Cost.✓
- Adequate wind.✓

(Any 2) (2)

3.1.2 **TWO advantages and TWO disadvantages of wind turbines.**

Advantages	Disadvantages
• No fuel costs.✓ • Low maintenance costs.✓ • No clean-up costs.✓ • No carbon tax costs.✓ • No air pollution.✓ • Wind is a renewable energy source.✓ • Wind energy is much less expensive than other sources of energy.✓ • Wind turbines are a great resource to generate energy in remote locations.✓	• High initial cost.✓ • Winds unreliability factor.✓ • In many areas, the winds strength is too low.✓ • Wind turbines generally produce allot less electricity than the average fossil fuelled power station/ Environmental impact✓ • Costly to surrounding wildlife during the build process.✓ • The noise pollution.✓ • Protests and/or petitions usually confront any proposed wind farm development.✓

(Any 2)

(Any 2)

(4)

3.1.3 **The use of wind energy during windless days and motivation.**

Yes✓, by making use of batteries.✓

(2)

3.2 **The manufacturing process of biodiesel.**

Making use of vegetable oils✓, algae or animal fats, involves the base-catalysed transesterification of fatty acids with methanol or ethanol to give the corresponding methyl esters or ethyl esters.✓

(2)

TWO plants from which ethanol can be made.

3.3 Maize✓, sorghum✓, potatoes✓, wheat✓, sugar-cane✓, cornstalks✓, fruit✓ or vegetable waste.✓

(Any 2)

(2)

3.4 3.4.1 Explanation of solar power.

Electricity generated from the levels of natural energy contained within the sun's rays.✓ This solar energy is commonly known as solar radiation, and contains a significant amount of energy we are able to harness.✓ The sun's rays contain photons which means they are able to carry an electrical charge.✓ (Any 2)

(2)**3.4.2 The working of the solar hot-water system.**

- The sun heats the water in the glass tubes; the heated water rises to the highest point in the system/ Thermosiphon action.✓
- The heated water enters the geyser through a closed copper pipe network that runs through the geyser.✓
- The hot water inside the copper pipes heats up the cold water inside the geyser.✓
- The cooled water flows downwards back to the solar tubes where it is reheated.✓

(4)**3.4.3 TWO reasons why solar hot-water system is regarded as a popular choice to heat water.**

- Environmentally friendly energy source/No pollution.✓
- Installation is relatively cheap and simple.✓
- Solar power is limitless.✓
- Transition losses are limited.✓
- Does not use a lot of space.✓

(Any 2)**(2)
[20]**

QUESTION 4: SKILLS AND CONSTRUCTION PROCESSES

- 4.1 4.1.1 **TWO surfaces that the earth clamp can be attached to, to complete the circuit.**
- Either be clipped directly to the piece of metal✓or onto a metal welding table.✓ (2)
- 4.1.2 **TWO causes of a lack of contact between the welding clamp and the work piece.**
- Loose wiring.✓
 - Dirty metal.✓
 - Paint.✓
 - Grease or oil.✓
 - Rust.✓
- (Any 2) (2)
- 4.1.3 **Factors to remember when welding thinner metals with the MIG welding machine.**
- Lower the power/current setting✓to prevent melting through the metal.✓ (2)
- 4.2 **Pipe-welding positions A and B.**
- A** - Horizontal movable position.✓
- B** - Vertical position.✓ (2)
- 4.3 **Description of the overhead arc welding position.**
- Set an arc as short as possible✓and weld a number of runs without any lateral movement.✓ When molten metal starts dripping, the amperage should be reduced slightly.✓ Move electrode slightly faster✓and hold electrode in same position as in relation to base metal.✓ (Any 4) (4)
- 4.4 4.4.1 **Identification and description of the welding process.**
- Hard-facing✓ (1)
- 4.4.2 **THREE steps that must followed before the welding process starts.**
- Area to be covered must be free of corrosion, grease or foreign substances by grinding it away.✓ Do not work at too high amperage.✓ Use a longer arc.✓ Use a special welding electrode.✓ (Any 3) (3)

- 4.5 4.5.1 **TWO factors that cause metal to distort during the welding process.**
- Amount of welding.✓
 - Number of welding runs.✓
 - Degree of resistance.✓
 - Original state or condition of parts that must be welded.✓
 - Welding procedure.✓
 - Heat input.✓
 - Restraint.✓
- (Any 2) (2)
- 4.5.2 **TWO ways in which welding heat distortion can be controlled.**
- Pre-setting.✓
 - Welding of patch work.✓
 - Clamping.✓
 - Spot welding.✓
- (Any 2) (2)
- 4.6 4.6.1 **THREE safety regulations being ignored by the operator.**
- Not wearing safety glasses.✓
 - Not wearing safety gloves.✓
 - Gas cylinders not fixed.✓
 - Liquid spilled on floor.✓
 - Fire hose next to working table.✓
 - Working table on liquid cans.✓
 - Working bench not cleared.✓
- (Any 3) (3)
- (Accept any visible safety rules violation.)
- 4.6.2 **The possible outcome if one of the gas cylinders were to topple over.**
- Someone might get hurt or die if the cylinder were to land on them.✓
 - Main valve might break off and cause the cylinder to become missile like and cause extreme damage.✓
- (2)
- 4.6.3 **TWO types of metal that can be cut with an oxy-acetylene welding set.**
- Mild steel.✓
 - Cast iron.✓
 - Stainless steel.✓
- (Any 2) (2)
- 4.7 4.7.1 **Name of the process.**
- Galvanizing of metal.✓
- (1)

4.7.2 **Reason why it is NOT recommended to weld directly on the zinc coating.**

It gives off poisonous smoke.✓

(1)

4.8 4.8.1 **Identification of plasma cutting machines A and B.**

A - Manual plasma cutting machine.✓

B - Automatic / computer generated plasma cutting machine/ CNC.✓

(2)

4.8.2 **TWO factors to consider when choosing the correct plasma cutting machine.**

- Analyse the minimum and maximum thickness of the metal to be cut.✓
- The type of cutting to be performed / fixed or moveable work.✓
- The voltage must be high enough to emit the correct amount of power for the work to be completed.✓

(Any 2)

(2)

4.8.3 **What to keep in mind when operating a plasma cutting machine in the overhead cutting position.**

Molten metal tends to drip down ✓ and might cause serious burns to the body.✓

(2)

[35]

QUESTION 5: TOOLS, IMPLEMENTS AND EQUIPMENT**Start this question on a NEW page.**

- 5.1 5.1.1 **TWO advantages of using a round baling machine on a farm.**
- One-man operation✓
 - Low rope consumption✓
 - Simplistic working✓
 - Can continue baling until rain starts✓
 - Roofed storage not necessary✓
- (Any 2) (2)
- 5.1.2 **FOUR procedures to follow when a baler is being prepared after not being used for a long time.**
- All grease points must be well greased.✓
 - The correct tension must be set for all belts or chains.✓
 - Check that all parts are functioning correctly by operating it slowly.✓
 - Replace all worn parts immediately.✓
 - Service according to manufacturer's specifications.✓
 - Lift up all dust release guards.✓
 - Check that binding timing is set correctly.✓
 - Check that machines magazine has sufficient rope.✓
 - Manufacturers specifications. ✓
- (Any 4) (4)
- 5.1.3 **TWO types of expenses of a round baler and give ONE example of each.**
- Fixed expenses.✓** (1)
- Depreciation.✓
 - Interest on invested capital.✓
 - Incidental expenses.✓
 - Financing cost✓
 - Licensing. ✓
- (Any 1) (1)
- Running expenses.✓** (1)
- Repairs.✓
 - Grease.✓
 - Labour.✓
 - Supervising expenses.✓
- (Any 1) (1)
- 5.1.4 **TWO functions of a shear bolt used in the drive system of a baling machine.**
- Prevents heavy objects from being taken into the baler.✓
 - Protects the pick-up if it is impeded by anything.✓
 - Protects the auger if it becomes overloaded.✓
- (Any 2) (2)

5.1.5 **THREE safety measures that should be considered when attaching the drive shaft of the baling machine to the power take-off shaft of the tractor.**

- Check that the shaft is as deep as possible and locked in place.✓
 - Check that the driving shaft guard is present.✓
 - Ensure that the tractor, PTO shaft and driving shaft of implement are aligned.✓
 - All grease points greased.✓
 - Engine of tractor must be switched off.✓
 - PTO shaft disengaged.✓
 - No damaged parts.✓
 - No vibrations in shaft.✓
- (Any 3) (3)

5.2 5.2.1 **FOUR safety hazards to consider when working with the cutting machine.**

- Flying debris/object can cause injuries/damage.✓
 - PTO must be guarded to prevent injuries.✓
 - Do not operate the machine on a steep slope.✓
 - Machine must be adjusted to the correct height.✓
- (3)

5.2.2 **TWO advantages of installing a bearing on the drive system of a cutting machine.**

- Bearings increase efficiency.✓
 - Bearings reduce friction and wear.✓
 - Allows extended use at high speeds.✓
 - Avoids overheating and premature failure of a drive system.✓
- (Any 2) (2)

5.3 5.3.1 **THREE points that must be considered by a farmer before purchasing a hammer mill.**

- Sturdy construction.✓
 - Replaceable wearing parts.✓
 - Rotor housing should close tightly.✓
 - Size of the hopper feed aperture/opening✓
- (Any 3) (3)

5.3.2 **FOUR points to consider when installing a hammer mill.**

- It must be installed in such a way that it can be handled with ease.✓
 - Placed in a well-ventilated area.✓
 - Fodder must be offloaded next to the hopper.✓
 - Mill should be placed near the feed mixer✓
- (4)

5.3.3 **ONE function for each of the following components of a hammer mill.**

- (a) **Hopper** - Facilitate the proses of feeding.✓ (1)
(b) **Cyclone** - Separate the grounded material effectively from the air.✓ (1)

5.4 **Calculation of the paint required.**

Paint required = $9 \text{ m}^2 / 3 \text{ m}^2$ ✓ (covering capability per litre)
= 3 litres needed✓
= 2 coats✓ X 3 litres per coat✓
= 6 litres of paint✓ (5)

5.5 **Solutions for the different problems associated with the hitching of an implement.**

- 5.5.1 Add counter-weights at the front of tractor.✓ (1)
5.5.2 Adjust the levelling box.✓ (1)
5.5.3 Extend the top link of the tractor.✓ (1)

5.6 **THREE factors that have an influence on the depth control system of a tractor.**

- Ploughing depth.✓
 - Soil resistance.✓
 - Forward speed of the tractor.✓
 - Draft control sensitivity.✓
- (Any 3) (3)
[40]

QUESTION 6: WATER MANAGEMENT

Start this question on a NEW page.

- 6.1 6.1.1 **FOUR reasons for the use of irrigation sprinklers instead of flood irrigation.**
- When water supply is low.✓
 - A high steep gradient leads to erosion.✓
 - Infiltration tempo not constant.✓
 - Type of soil.✓
 - Type of crop.✓
 - Water efficiency. ✓
 - Labour saving. ✓
 - Fertigation.✓
- (Any 4) (4)
- 6.1.2 **FOUR primary parts of sprinkler irrigation system.**
- Valves.✓
 - Filter.✓
 - Pump.✓
 - Pipes.✓
 - Sprinklers.✓
 - Pressure regulator
- (Any 4) (4)
- 6.1.3 **THREE types of irrigation systems that use sprinklers.**
- Centre pivot Irrigation system.✓
 - Travelling sprinkler gun or hose reel irrigation machine.✓
 - Lateral move (side roll, wheel line) irrigation.✓
 - Portable 'Hand move' sprinkler system.✓
- (Any 3) (3)
- 6.2 6.2.1 **Reason why a farmer uses drainage systems in a field.**
- Drainage systems remove excess or free water from the upper layers of waterlogged soil/Subsurface drainage✓
- (1)
- 6.2.2 **FOUR types of drains used to drain excess water from fields.**
- French drain.✓
 - Channel drain.✓
 - Slope drains.✓
 - Drainage ditch.✓
- (Any 4) (4)
- 6.3 6.3.1 **Function of the sieve A on the outlet side of the septic tank.**
- Prevents any solids to go out of the outlet.✓
 - Prevents foreign objects entering the septic tank from the outlet side.✓
- (2)

6.3.2 Description of the working of a septic tank from the moment the water enters the inlet side of the tank to the moment the water exits the outlet side.

- Sewage is broken down by anaerobic bacteria in the first tank.✓
- Very little solids remain when the watery sewerage flows to the second tank.✓
- Only liquid sewage remains and drains away through the outlet pipe or stone trench✓

(3)

6.4 Table regarding different irrigation systems.

6.4.1 Centre pivot irrigation✓ (1)

6.4.2 PVC pipes with plastic or brass sprinklers✓ (1)

6.4.3 Sprinklers can irrigate areas such as small farms, parks and pastures✓ (1)

6.4.4 Lateral wheel line irrigation✓ (1)

6.5 The functions for EACH part of an irrigation system.

6.5.1 Sand filter

Prevents blockage of sprinkler nozzles / filters the water.✓ (1)

6.5.2 Irrigation timer

To control the irrigation system.✓ (1)

6.5.3 Neutron moisture probe

Measure the moisture content in the soil.✓ (1)

6.5.4 Solenoid valve

Controlling the water flowing through the system.✓ (1)

6.5.5 One-way valve

Prevent water from flowing back to the pipe.✓ (1)

[30]

TOTAL SECTION B: 160
GRAND TOTAL: 200